

5.6 Astrophysics and cosmology

This topic covers the physical interpretation of astronomical observations, the formation and evolution of stars, and the history and future of the universe.

Students will be assessed on their ability to:	Suggested experiments
126 use the expression $F = Gm_1m_2/r^2$	
127 derive and use the expression $g = -Gm/r^2$ for the gravitational field due to a point mass	
128 recall similarities and differences between electric and gravitational fields	
129 recognise and use the expression relating flux, luminosity and distance $F = L/4\pi d^2$ application to standard candles	
130 explain how distances can be determined using trigonometric parallax and by measurements on radiation flux received from objects of known luminosity (standard candles)	
131 recognise and use a simple Hertzsprung-Russell diagram to relate luminosity and temperature. Use this diagram to explain the life cycle of stars	
132 recognise and use the expression $L = \sigma T^4 \times \text{surface area}$, (for a sphere $L = 4\pi r^2 \sigma T^4$) (Stefan-Boltzmann law) for black body radiators	
133 recognise and use the expression: $\lambda_{\text{max}} T = 2.898 \times 10^{-3} \text{ m K}$ (Wien's law) for black body radiators	

Students will be assessed on their ability to:	Suggested experiments
134 recognise and use the expressions $z = \Delta\lambda/\lambda \approx \Delta f/f \approx v/c$ for a source of electromagnetic radiation moving relative to an observer and $v = H_0 d$ for objects at cosmological distances	
135 be aware of the controversy over the age and ultimate fate of the universe associated with the value of the Hubble Constant and the possible existence of dark matter	
136 explain the concept of nuclear binding energy, and recognise and use the expression $\Delta E = c^2 \Delta m$ and use the non SI atomic mass unit (u) in calculations of nuclear mass (including mass deficit) and energy	
137 describe the processes of nuclear fusion and fission	
138 explain the mechanism of nuclear fusion and the need for high densities of matter and high temperatures to bring it about and maintain it	